

Nonlinear MPC for a Planar Biped using Contact-Constrained Trajectory Optimization

Guangxuan Xu

University of Michigan

Department of Biomedical Engineering

Department of Robotics

Ann Arbor, USA

eggsyxu@umich.edu

Abstract—In this report, we developed a nonlinear model predictive control framework for a 5-link biped robot with a contact-constrained trajectory optimization method to generate squatting, walking and jumping motion. I was primarily responsible for the jumping part, so I will focus on the implementation of trajectory optimization for jump motions.

Index Terms—Mpc, Biped Robot, Trajectory Optimization

I. INTRODUCTION

Most of the time we choose either full-body dynamics or single point model to do the modeling of humanoid robots [1] [2]. It might be too complex or too simple to do either of it [3] [4] [5]. So, the purpose of this project is to find the balance of it, even if there are works previous than us [6] [7] [8]. To reduce computational complexity during jumping motion, the robot is modeled using 5-link model during the take-off phase, while a single-rigid-body (SRB) model is used during the flight and landing phases.

II. IMPLEMENTATION AND RESULTS

In this project, three implementation frameworks were used: a 3-link MATLAB model, a 5-link MATLAB model, and a MuJoCo simulation model. The implementation is structured into several components, including modeling, kinematics and dynamics formulation, trajectory optimization (TO), and model predictive control (MPC). My primary responsibilities in this project were the setup of the MuJoCo simulation environment and the implementation of the trajectory optimization (TO) pipeline. Therefore, this report will primarily focus on the five-link model and the jumping motion. The overall pipeline shows in Algorithm 1.

A. Modeling

The five-link robot model consists of three main body links, each with a length of 0.4 m, with the joint configuration illustrated in Fig. 1.

$$q = [x, y, \theta, q_1, q_2, q_3, q_4]^\top \quad (1)$$

where x and y denote the torso position, θ is the torso orientation, and q_1 – q_4 represent the four joint angles.

Algorithm 1 TO and MPC for Planar Biped Jumping

Input: current state, current phase, landing state

Output: joint torques

- 1: Initialize phase lengths N_t, N_f, N_l , time steps dt_t, dt_f, dt_l
 - 2: Initialize phase sequence: Takeoff $\rightarrow$ Flight $\rightarrow$ Landing
 - 3: Offline Trajectory Optimization
 - 4: Takeoff: five-link dynamics with Forward Euler
 - 5: Flight, Landing: SRB dynamics with Forward Euler
 - 6: Add object function and constraints
 - 7: Solve NLP
 - 8: Extract reference trajectories $x_c^{\text{ref}}(k), q_j^{\text{ref}}(k)$
 - 9: **while** jump not finished **do**
 - 10: Measure $x_c, q_j, \dot{q}_j$
 - 11: Build MPC state x_k and reference x_k^{ref} from X^*
 - 12: Solve MPC:
 - 13: $x_{k+1} = \hat{A}_{d,k}x_k + \hat{B}_{d,k}u_k$
 - 14: subject to friction constraints
 - 15: Obtain optimal ground reaction wrench u_k^*
 - 16: Compute joint torque:
 - 17: $\tau_j = K_p(q_j^{\text{ref}} - q_j) + K_d(\dot{q}_j^{\text{ref}} - \dot{q}_j) + J_c^\top u_k^*$
 - 18: Update phase
 - 19: **end while**
-

B. Kinematics and Dynamics

The kinematic model is shown above, while different dynamic models are used for different phases of the jumping motion.

For take-off phase, we have the state of $X = [q, \dot{q}]^\top$, and the dynamic equation is:

$$\mathbf{H}(q)\ddot{q} + \mathbf{C}(q, \dot{q}) = \Gamma + J_i(q)^\top \lambda_i \quad (2)$$

$$J\ddot{q} + \dot{J}\dot{q} = 0 \quad (3)$$

where $\mathbf{H}(q)$ is the inertia matrix, $\mathbf{C}(q, \dot{q})$ is the Coriolis/centrifugal term, Γ is the actuation torque vector, $J_i(q)$ is the Jacobian of contact point i , and λ_i is the corresponding contact force.

During the flight and landing phase, the state is $X = [p_c, \theta, \dot{p}_c, \dot{\theta}]^\top$, where p_c is the COM position and θ the torso orientation.

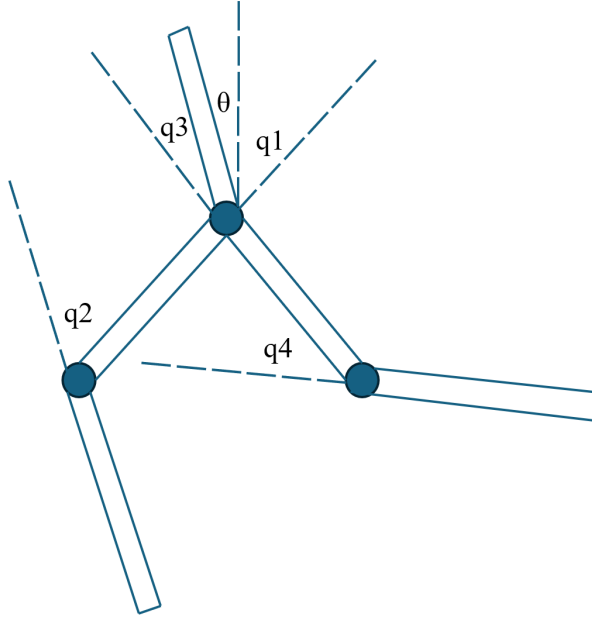


Fig. 1. Five-link planar biped model.

And the dynamic equation for SRB should be:

$$m_0(\ddot{p}_c^{2D} + g) = F \quad (5)$$

$$\frac{d}{dt}(I_c \dot{\theta}) = (p_f - p_c^{2D}) \times F + M \quad (6)$$

where F is the 2D ground reaction force at the contact point, and M is the ground reaction moment in the x - z plane. For flight phase, since there is no contact with the ground, there supposed to be no contact force. But we involved a virtual force here, to smooth the overall dynamics conditions.

C. Trajectory Optimization

I will primarily describe the trajectory optimization (TO) implementation for the jumping motion in this report.

First, the time sequence of the jumping motion is shown in Fig. 2.

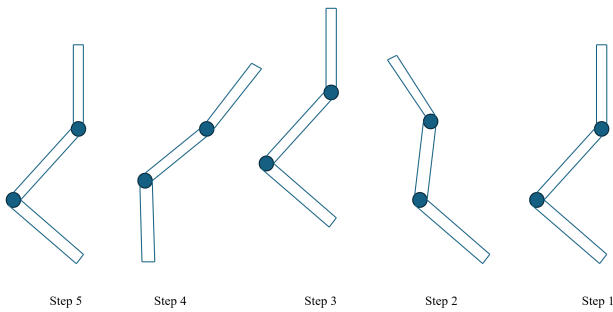


Fig. 2. Jumping sequence of the five-link planar biped.

With the phase-dependent models introduced earlier, this section presents the trajectory optimization formulation used

to generate jumping motions for the five-link planar biped. This model determines take-off, flight and landing time step previously with different frequency.

This TO problem with N steps is formulated as:

$$\min_X \sum_{i=1}^N (\|q_i - q_0\|_{Q_0}^2 + \|\dot{q}_i\|_{Q_1}^2 + \|\tau_i\|_{Q_2}^2), \quad (4)$$

subject to the following constraints.

1) *Physical Constraints:*

$$q_{\min} \leq q_i \leq q_{\max}, \quad (5)$$

$$\dot{q}_{\min} \leq \dot{q}_i \leq \dot{q}_{\max}, \quad (6)$$

$$\tau_{\min} \leq \tau_i \leq \tau_{\max}. \quad (7)$$

2) *Initial and Final Conditions:*

$$x_{c,0} = x_{IC}, \quad (8)$$

$$x_{c,N} = x_{FC}. \quad (9)$$

3) *Take-off Phase:* $i = 0 : N_t - 1$:

$$\ddot{q}_i = M^{-1}(-C + \tau_i + G) \quad (10)$$

$$q_{i+1} = q_i + \dot{q}_i dt_t \quad (11)$$

$$\dot{q}_{i+1} = \dot{q}_i + \ddot{q}_i dt_t - l_h \leq p_{c,x}(q_i) \leq l_t. \quad (12)$$

4) *Flight Phase:* $i = N_t : N_t + N_f - 1$:

$$x_{c,i+1} = x_{c,i} + \dot{x}_{c,i} dt_f p_{toe,y}(q_i) > h_{\text{terrain}} \quad (13)$$

5) *Landing Phase:* $i = N_t + N_f : N - 1$:

$$-\mu F_{y,i} \leq F_{x,i} \leq \mu F_{y,i}, \quad (14)$$

In summary, this trajectory optimization has a fixed time sequence with pre-ordered planning, applied by a simple forward-Euler iteration during the optimization process. This design ensures solver searches for optimal dynamics within a human-intuitive motion sequence, rather than producing unnatural behaviors. As a result, the generated trajectories remain smooth, physically consistent, and well suited for execution by the MPC.

D. Model Predictive Control

Once the TO data was generated offline, the subsequent variable-frequency MPC operates together with a joint PD controller to track the reference trajectories x_c^{ref} and q_j^{ref} , which are obtained directly from the optimization solution X . The MPC will track the sequence with different frequency according to the phases and the sampling step length dt will contact to it to ensure the controller have the same resolution as TO did.

For this linearize discrete-time state space system, we have:

$$x_{k+1} = \hat{A}_{d,k} x_k + \hat{B}_{d,k} u_k,$$

The control input u consists of the 3D ground reaction forces and moments for both feet is:

$$u = [F_0; F_1; M_0; M_1].$$

The MPC problem over a finite horizon h will be:

$$\min_{x,u} \sum_{i=0}^{h-1} (\|x_k - x_k^{\text{ref}}\|_{R_0}^2 + \|u_k\|_{R_1}^2), \quad (13)$$

$$\begin{aligned} \text{Friction:} \quad & -\mu F_{z,n} \leq F_{x,n} \leq \mu F_{z,n}, \\ & -\mu F_{z,n} \leq F_{y,n} \leq \mu F_{z,n}, \end{aligned} \quad (14b)$$

$$\text{Force limit:} \quad 0 \leq F_{z,n} \leq F_{\max}, \quad (14c)$$

$$\text{Moment (X-axis):} \quad BM_{x,n} = 0, \quad (14d)$$

$$\text{Line-foot constraint:} \quad -l_h F_{z,n} \leq M_{y,n} \leq l_t F_{z,n}. \quad (14e)$$

The real-time jumping control consists of both a variable-frequency MPC for tracking the CoM trajectory and a joint PD controller for tracking the joint trajectory simultaneously. The joint torque command is computed as

$$\tau_j = K_p(q_j^{\text{ref}} - q_j) + K_d(\dot{q}_j^{\text{ref}} - \dot{q}_j) + J_c^\top u, \quad (15)$$

Fig. 3 illustrates the control performance for a single jump.

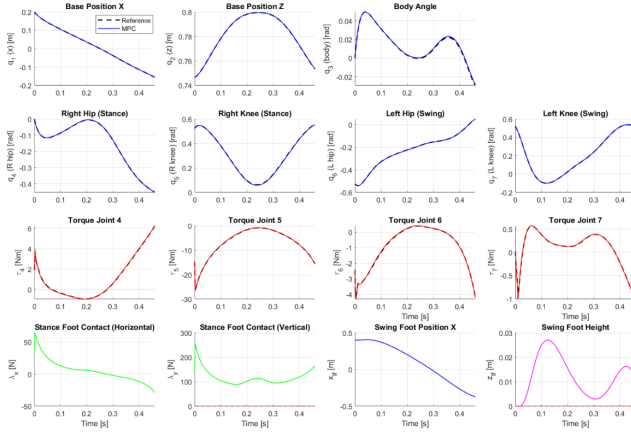


Fig. 3. MPC performance during a single jump

CoM trajectory is tracked smoothly by MPC both in X and Z axis, the body pitch remains still during the flight. Two legs' angles and torque work just fine as shown.

At the beginning of the simulation, the nonlinear MPC problem is solved with a poor initial guess leading to a long time (nearly 3 seconds). We thought it might be the Simpson's rule that make this MPC working so slow, so we change it to forward-Euler method. As a result, this nonlinear optimizer start working rapidly and each step can stabilize at approximately 0.1 seconds shown in Fig. 4.

E. MuJoCo Setup

To validate our trajectory optimization and control framework, I implement a simplified planar biped model in MuJoCo, as shown in Fig. 5. The model consists of a rigid torso and two legs, each represented by two cylindrical links. The upper leg (blue) and lower leg (green) are connected through a single revolute knee joint, while the hip joint connects each leg to the floating-base torso. No ankle joint is modeled, and the feet are

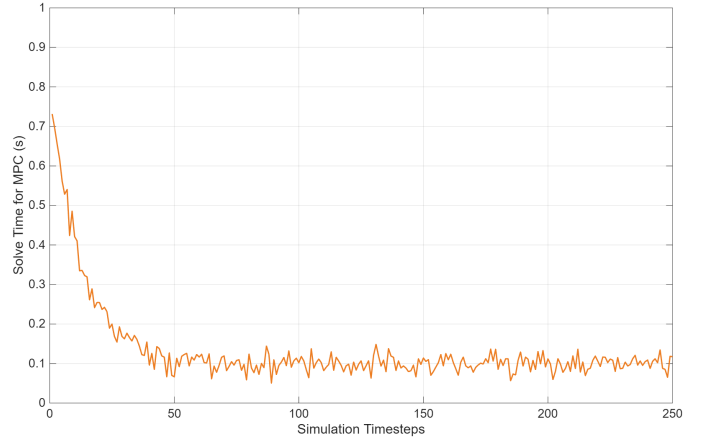


Fig. 4. MPC running time

represented as simple point contacts to reduce the complexity.

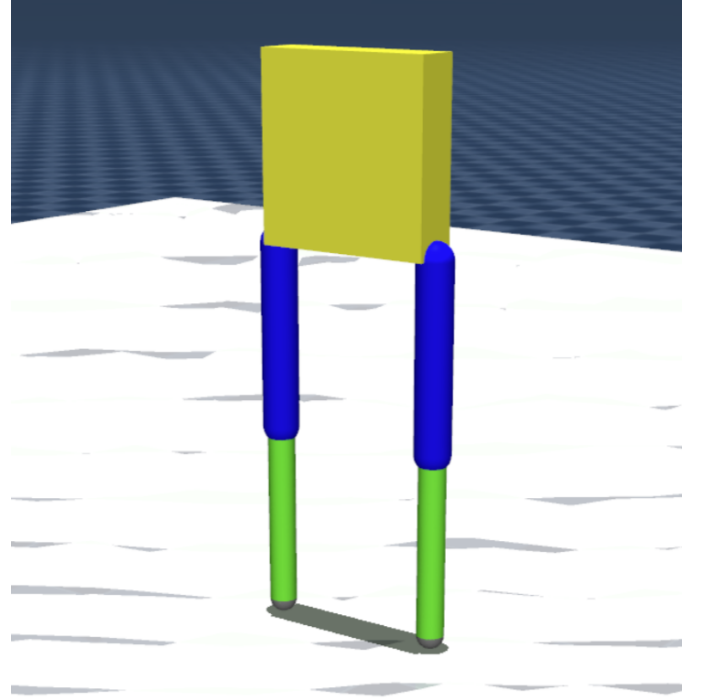


Fig. 5. MuJoCo five-link model

III. ACKNOWLEDGMENT

I would like to express my sincere gratitude to my teammates, Yutong and Vansh, for working with me throughout this project. Their support and collaboration were essential to completing the core components of our system.

The only regret for this project is that I was not able to finish the MuJoCo jumping implementation. Shortly before Thanksgiving, I was involved in a car accident and had to spend several days in the emergency department followed by a period of recovery at home. As a result, I did not have

enough time to complete the simulation module as originally planned.

This semester has been quite eventful, and I even had to return to China briefly, which made it difficult for me to attend all in-person classes. Despite this, I was able to follow the online lectures and complete all the exercises, and I learned a great deal from the course. I am sincerely grateful to Professor Yanran, Yulun, and my teammates for their support, understanding, and collaboration throughout the semester.

REFERENCES

- [1] S. Kuindersma, R. Deits, M. Fallon, A. Valenzuela, H. Dai, F. Permenter, T. Koolen, P. Marion, and R. Tedrake, "The MIT Humanoid Robot: Design, Motion Planning, and Control for Acrobatic Behaviors," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, pp. 345–352, 2015.
- [2] J. Di Carlo, P. M. Wensing, B. Katz, G. Bledt, and S. Kim, "Online Trajectory Optimization for Dynamic Aerial Motions of a Quadruped Robot," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 1–7, 2018.
- [3] T. Koolen, H. Dai, J. Rebula, L. Johnson, J. Pratt, and R. Tedrake, "Implementing Regularized Predictive Control for Simultaneous Real-Time Footstep and Ground Reaction Force Optimization," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, pp. 1–8, 2016.
- [4] M. Posa, S. Kuindersma, and R. Tedrake, "Whole-body Motion Planning with Centroidal Dynamics and Full Kinematics," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, pp. 295–302, 2016.
- [5] A. Herzog, N. Rotella, S. Mason, et al., "Balancing Experiments on a Torque-Controlled Humanoid with Hierarchical Inverse Dynamics," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 981–988, 2014.
- [6] A. D. Ames, K. Galloway, K. Sreenath, and J. W. Grizzle, "Rapidly Exponentially Stabilizing Control Lyapunov Functions and Hybrid Zero Dynamics," in *Proc. IEEE Conf. Decision and Control (CDC)*, pp. 2922–2929, 2012.
- [7] M. Posa, C. Cantu, and R. Tedrake, "A Direct Method for Trajectory Optimization of Rigid Bodies Through Contact," in *Proc. Robotics: Science and Systems (RSS)*, 2014.
- [8] D. Orin and A. Goswami, "Centroidal Momentum Matrix of a Humanoid Robot: Structure and Properties," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 653–660, 2008.